EXPERIMENT 4

Academic Year: 2025-26 **Year and Sem: BEVIII**
Subject: Recommendation Systems Lab **Course Code: CSDOL8022**

Aim: To implement knowledge base recommendation system using Python
Theory: A Knowledge-Based Recommendation System (KBRS) is a type of recommendation system that provides suggestions based on explicit knowledge about products and user requirements. This knowledge is represented through rules, constraints, and logical relationships between user preferences and product attributes.
Unlike collaborative filtering or content-based recommendation systems, knowledge-based systems do not rely on historical user data such as ratings or purchase history. Instead, they use domain knowledge and predefined rules to match user requirements with appropriate items.
In this system, recommendations are generated by comparing user inputs (preferences) with item attributes stored in a dataset.
For example, in a smartphone recommendation system, the dataset may contain attributes such as:

- Product Name
- Brand
- Price
- RAM
- Storage
- Camera Quality
- Battery Capacity

If a user specifies preferences like brand = Apple, price < \$1000, RAM ≥ 6 GB, the system filters the dataset and recommends products that satisfy these conditions.
Knowledge-based recommendation systems are particularly useful when:

- There is no historical user data available.
- The system contains new products.
- Recommendations depend on specific user requirements.

Key Characteristics of Knowledge-Based Recommendation Systems
1. Explicit Knowledge Representation
Knowledge-based systems rely on **explicit knowledge about items and user preferences**. This knowledge is typically stored as **structured information**, including product attributes, rules, or domain-specific facts.
2. Rule-Based Recommendation
These systems commonly use **if–then rules** to represent the relationship between user
3. Independence from Historical Data
Knowledge-based recommendation systems **do not require past user interaction data**. This makes them useful when:

- The system has **new users with no previous activity**
- **New products** are added to the system

This approach helps address the **cold start problem** commonly found in other recommendation techniques.

4. User-Driven Preferences

In this approach, users **directly specify their requirements**, which are used to generate recommendations.

5. Use of Domain Knowledge

Knowledge-based systems incorporate **domain-specific knowledge** into the recommendation process. This knowledge may include **product specifications, expert opinions, or technical constraints**, which help improve the accuracy of recommendations.

Algorithm

Step 1: Start the program.
Step 2: Import required libraries such as Pandas.
Step 3: Create or load the dataset containing product information (product id, name, brand, price, RAM, etc.).
Step 4: Accept user preferences such as:

- Preferred brand
- Maximum price
- Minimum RAM

Step 5: Apply filtering conditions to match user preferences with product attributes.
Step 6: Select products that satisfy all the specified constraints.
Step 7: Display the list of recommended products.
Step 8: Stop the program.

Conclusion

The experiment demonstrated how knowledge-based recommendation systems generate suggestions using explicit rules and user requirements. Unlike collaborative filtering methods, this approach does not require historical user data and can effectively recommend items based on constraints and domain knowledge.